Expedited Noise Spectroscopy Of Transmon Qubit

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Noise in transmon qubits

Sources	Arises from	Noise mechanisms	S(ω)
Charge Noise	Charge fluctuators present in the defects or charge traps that reside in interfacial dielectrics, the junction tunnel barrier, and in the substrate itself.	Dominant: Longitudinal (T_1), Sometimes low frequency (T_ϕ)	At low frequency $\propto 1/\omega$, at high frequency $\propto \omega$
Magnetic Flux Noise	Stochastic flipping of qubits that reside on the surfaces of the superconductors, resulting in random fluctuations of effective-$\vec{B}$ that biases flux-tunable qubits.	Longitudinal (T_1), Transverse (T_ϕ)	$\propto 1/\omega$
Photon Number Fluctuations	In resonator, residual microwave fields in the cavity have photon number fluctuations.	Transverse (T_ϕ)	$\propto \frac{k}{\omega^2+k^2}$; k : resonator decay rate
Quasiparticles	Unpaired electrons tunneling through a qubit junction.	Longitudinal (T_1), Transverse (T_ϕ)	$\propto \frac{N_{qp}\tau_{qp}}{\omega^2+\tau_{qp}^2}$; (N_{qp}, τ_{qp}) : quasiparticle (population, lifetime)
Local Two-Level Systems (TLS)	Electric dipole moments resonantly absorb energy from the oscillating $\vec{E}$ of the qubit mode, and efficiently dissipate it into the phonon or quasiparticle bath.	Longitudinal (T_1), Transverse (T_ϕ)	Low freq.: White noise, $1/\omega$; High freq.: Lorentzian type

$$\rho_{noisy} = \begin{pmatrix} 1+(|\alpha|^2-1)e^{-\Gamma_1 t} & \alpha\beta^* e^{-\frac{\Gamma_1}{2}t}e^{-\Gamma_\phi t} \\ \alpha^*\beta e^{-\frac{\Gamma_1}{2}t}e^{-\Gamma_\phi t} & |\beta|^2 e^{-\Gamma_1 t} \end{pmatrix}$$

C(t) = e^(-χ(t))

Decoherence can be mitigated by switching the qubits at some frequency

Ex:

CPMG-32 sequence

- Make a superposition state using a pi/2 pulse
- Sequence of 32 pi pulse followed by equal amounts of free evolution time
- pi/2 pulse to rotate back to GND state. If any phase has accumulated due to noise, there will be an excited state component. Regular switching cancels phase drifting due to nose. So it acts as a filter on the noise spectrum

Filter function formulation and the inverse problem

$$H(t) = \frac{\hbar}{2}\left[\omega_0 + \delta\omega(t)\right]\sigma_z$$

$$\phi(t) = \int_0^t y(t_1)\,\delta\omega(t_1)\,dt_1$$

$$C(t) = \langle e^{i\phi(t)}\rangle = \exp\left(-\chi(t)\right), \quad \chi(t) = \frac{1}{2}\langle\phi^2(t)\rangle$$

$$\chi(t) = \frac{1}{2}\int_0^t dt_1 \int_0^t dt_2\, y(t_1)y(t_2)\,\langle\delta\omega(t_1)\delta\omega(t_2)\rangle$$

$$\langle\delta\omega(t_1)\delta\omega(t_2)\rangle = \int_{-\infty}^{\infty}\frac{d\omega}{\pi}S(\omega)\,e^{-i\omega(t_1-t_2)}$$

$$\chi(t) = \frac{1}{2}\int_0^{\infty} 2\,\frac{d\omega}{\pi}S(\omega)\left|\int_0^t y(t')e^{i\omega t'}dt'\right|^2$$

$$F(\omega t) = \omega^2\left|\int_0^t y(t')e^{i\omega t'}dt'\right|^2 \qquad (\tau = t'/t)$$

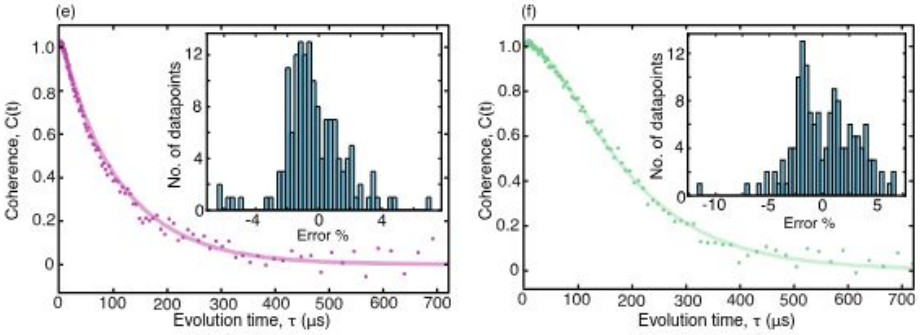
$$\chi(t) = \int_0^{\infty}\frac{d\omega}{\pi}\frac{S(\omega)}{\omega^2}F(\omega t)$$

Experimental procedure

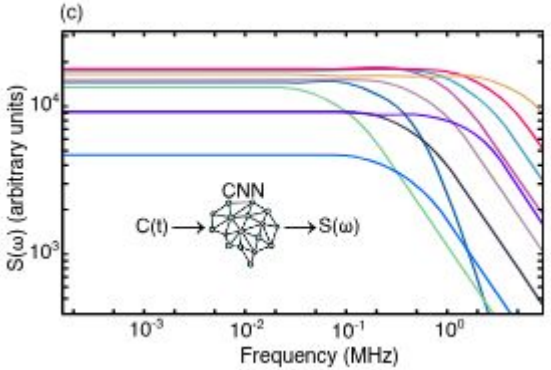
Machine - 127 qubit IBM Osaka ,superconducting qubits, Coherence times $T2 \approx 150-300 \mu s$

- $T \in [2\mu s, 720\mu s]$ is the range of total evolution times, 150 samples
- CPMG-32 pulses are applied for many shots across all times, to determine C(t), 4000 measurements per sample
- Predict noise spectrum using CNN, validate by reconstructing C(T) and comparing theory and experiment
- Find optimum pulse sequence for least noise
- Repeat 10 times to check for variation with time, each experiment takes ~ 10 minutes to run

Results



C(t) along with reconstructed C(t) for 2 different qubits

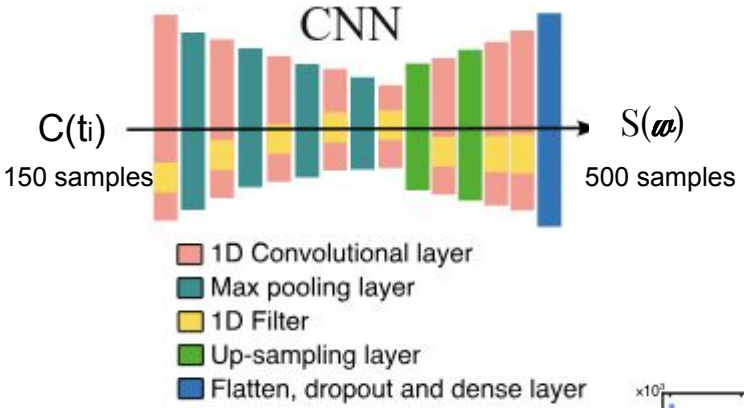


S(w) experimentally found for 10 different qubits

CNN for solving for S(w) from C(t)

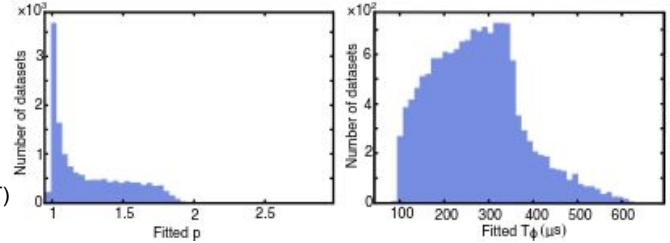
This is trained for CPMG-32, can be retrained for other protocols

Training pipeline

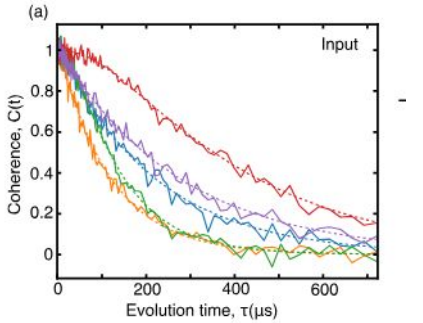


15 layers, 770k parameters

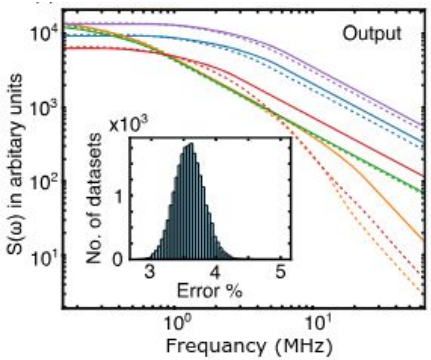
- Generate synthetic S(w) by combining white, lorentzian, 1/f and 1/f^2 noise. (2x10^5 samples)
- Calculate corresponding Coherence curves, add gaussian noise to simulate “experimental error”
- Test robustness with different amounts of noise, 3% gaussian gives best results
- Fit C(t) to e^(1(t/T)^p), discard unphysical C(t)
- Trained for 75 epochs



** Phase decay is e^(-t/T) for white noise, e^(-(t/T)^2) for 1/f noise

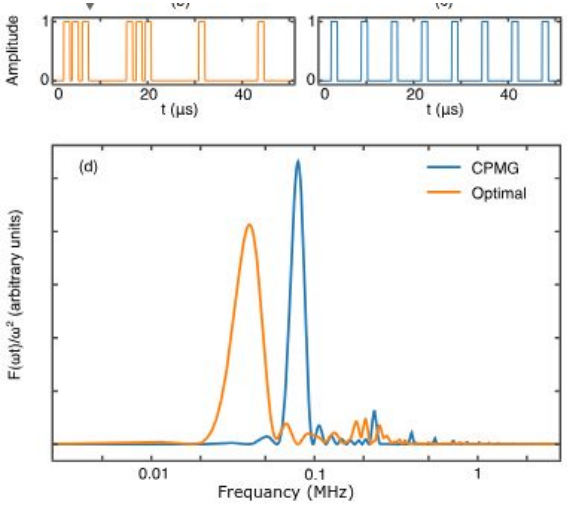


Samples of synthetic C(t) and corresponding spectrums, and its reconstruction



Optimizing pulse shape from Noise spectrum

Standard CPMG-8 and DD optimized pulses

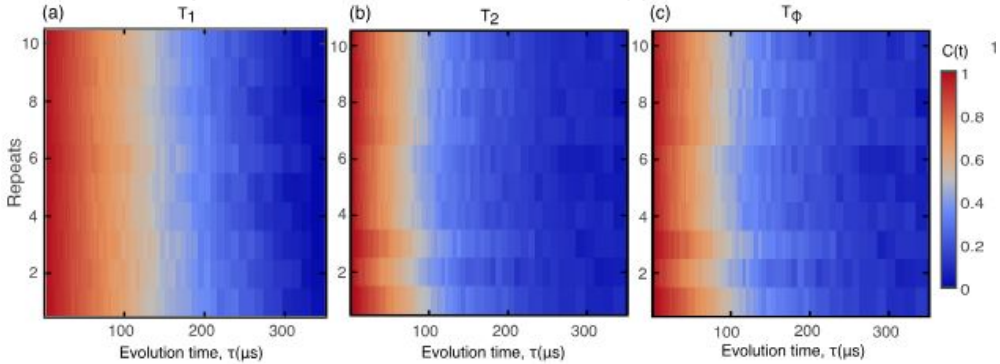


SciPy
Optimizer SLSQP used to optimize pulse

Having calculated S(w) previously, the optimum pulse for least phase noise is found.

- On IBM's qubits, as noise is very flat, optimizing pulse timing makes little difference, more significant with more complex spectrums.

The below plots show multiple repeats of the experiment, for IBM processor, the variation in C(t) is small



Thank You